

A Proposed MRTS-Augmented Bayesian Hierarchical Skew- t Model

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1 Goal

The goal of this note is to formulate a computationally efficient alternative to the spatial dependence component in the Morris skew- t hierarchical model. The main idea is to retain the original hierarchical structure for the mean, skew, and scale components, but replace the site-level Matérn correlation model with an MRTS-based low-rank spatial random effect representation. To avoid treating the full $K \times K$ covariance matrix $\mathbf{M}$ as a high-dimensional Bayesian parameter block, we propose to update $\mathbf{M}$ using the closed-form maximum likelihood estimators derived by Tzeng and Huang (2018), and then convert the resulting covariance matrix to a correlation matrix so that the original Morris separation between marginal scale and spatial dependence is preserved.

2 Original Morris Hierarchical Structure

Let $s_1, \dots, s_n \in D \subset \mathbb{R}^d$ denote the observed spatial locations and let $t = 1, \dots, T$ index replicates in time. Let $Y_t(s_i)$ denote the response at location s_i and replicate t . For each t , define the response vector

$$\mathbf{Y}_t = (Y_t(s_1), \dots, Y_t(s_n))'.$$

As in Morris et al., we retain the partition-based skew and scale structure. Let P_{tk} , $k = 1, \dots, K_t^{(p)}$, denote the spatial partitions at time t . For $s \in P_{tk}$, define

$$z_t(s) = z_{tk}, \quad \sigma_t^2(s) = \sigma_{tk}^2.$$

Let

$$\mathbf{z}_t = (z_t(s_1), \dots, z_t(s_n))', \quad \mathbf{D}_t = \text{diag}(\sigma_t(s_1), \dots, \sigma_t(s_n)).$$

Let $\mathbf{X}_t$ denote the $n \times p$ design matrix for replicate t , and let $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\lambda \in \mathbb{R}$ denote the regression and skewness parameters.

The original Morris model can be written in vector form as

$$\mathbf{Y}_t = \mathbf{X}_t \boldsymbol{\beta} + \lambda \mathbf{D}_t |\mathbf{z}_t| + \mathbf{D}_t \mathbf{v}_t, \quad t = 1, \dots, T, \quad (1)$$

where $|\mathbf{z}_t|$ is interpreted componentwise, and $\mathbf{v}_t$ is a zero-mean Gaussian spatial process with a correlation matrix determined by a Matérn family with a nugget-type parameter. In the original implementation, this correlation structure is governed by a small number of hyperparameters such as ρ , ν , and γ , but repeated MCMC updates still require repeated inversions of an $n \times n$ covariance matrix.

3 Proposed MRTS Spatial Layer

3.1 Basis Expansion

Let $f_1(\cdot), \dots, f_K(\cdot)$ denote prespecified MRTS basis functions, with $K \ll n$ in the intended applications. Define

$$\mathbf{f}(s) = (f_1(s), \dots, f_K(s))',$$

and let the $n \times K$ basis matrix be

$$\mathbf{F} = \begin{bmatrix} \mathbf{f}(s_1)' \\ \vdots \\ \mathbf{f}(s_n)' \end{bmatrix}.$$

Introduce a latent spatial random effect vector

$$\mathbf{u}_t = \mathbf{F}\boldsymbol{\eta}_t + \boldsymbol{\xi}_t, \quad \boldsymbol{\eta}_t \sim N_K(\mathbf{0}, \mathbf{M}), \quad \boldsymbol{\xi}_t \sim N_n(\mathbf{0}, \sigma_\xi^2 \mathbf{I}_n),$$

independently over t , where $\mathbf{M}$ is an unknown nonnegative-definite $K \times K$ matrix and $\sigma_\xi^2 \geq 0$ is a fine-scale variance parameter. Conditional on $(\mathbf{M}, \sigma_\xi^2)$,

$$\mathbf{u}_t \sim N_n(\mathbf{0}, \mathbf{G}), \quad \mathbf{G} = \mathbf{F}\mathbf{M}\mathbf{F}' + \sigma_\xi^2 \mathbf{I}_n.$$

3.2 Standardization to a Correlation Matrix

If $\mathbf{G}$ were used directly in (1), then the marginal variance of the spatial term would be controlled jointly by $\mathbf{G}$ and by $\mathbf{D}_t$, which would blur the original division of labor between the scale process and the spatial dependence process. To preserve the Morris interpretation, we convert $\mathbf{G}$ to a correlation matrix.

Define

$$\mathbf{A} = \text{diag}(\mathbf{G}), \quad \mathbf{R} = \mathbf{A}^{-1/2} \mathbf{G} \mathbf{A}^{-1/2}.$$

Then $\mathbf{R}$ is positive semidefinite with unit diagonal. We define the standardized latent spatial effect

$$\mathbf{v}_t = \mathbf{A}^{-1/2} \mathbf{u}_t, \quad \mathbf{v}_t \sim N_n(\mathbf{0}, \mathbf{R}).$$

The proposed hierarchical model is therefore

$$\mathbf{Y}_t = \mathbf{X}_t \boldsymbol{\beta} + \lambda \mathbf{D}_t |\mathbf{z}_t| + \mathbf{D}_t \mathbf{v}_t, \quad \mathbf{v}_t \sim N_n(\mathbf{0}, \mathbf{R}), \quad (2)$$

with

$$\mathbf{R} = \mathbf{A}^{-1/2} (\mathbf{F}\mathbf{M}\mathbf{F}' + \sigma_\xi^2 \mathbf{I}_n) \mathbf{A}^{-1/2}, \quad \mathbf{A} = \text{diag}(\mathbf{F}\mathbf{M}\mathbf{F}' + \sigma_\xi^2 \mathbf{I}_n).$$

3.3 Conditional Residual Representation

Given $(\boldsymbol{\beta}, \lambda, \mathbf{z}_t, \mathbf{D}_t)$, define the standardized residual vector

$$\mathbf{r}_t = \mathbf{D}_t^{-1} (\mathbf{Y}_t - \mathbf{X}_t \boldsymbol{\beta} - \lambda \mathbf{D}_t |\mathbf{z}_t|), \quad t = 1, \dots, T. \quad (3)$$

Under the proposed model,

$$\mathbf{r}_t = \mathbf{v}_t \sim N_n(\mathbf{0}, \mathbf{R}).$$

For the purpose of updating the low-rank covariance structure, we also consider the unstandardized low-rank covariance

$$\mathbf{G} = \mathbf{F}\mathbf{M}\mathbf{F}' + \sigma_\xi^2 \mathbf{I}_n,$$

and use the Tzeng–Huang closed-form estimators as a plug-in update for $(\mathbf{M}, \sigma_\xi^2)$ before converting $\mathbf{G}$ to $\mathbf{R}$.

4 Closed-Form Update for $\mathbf{M}$ and σ_ξ^2

4.1 Working Covariance Update

At a given MCMC iteration, treat the residual vectors $\mathbf{r}_1, \dots, \mathbf{r}_T$ in (3) as the current approximately Gaussian replicates used to update the spatial dependence block. Define the empirical residual covariance matrix

$$\mathbf{S} = \frac{1}{T} \sum_{t=1}^T \mathbf{r}_t \mathbf{r}_t'.$$

The spatial random effect model of Tzeng and Huang takes the form

$$\mathbf{z}_t = \mathbf{y}_t + \boldsymbol{\epsilon}_t, \quad \mathbf{y}_t = \mathbf{F} \boldsymbol{\eta}_t + \boldsymbol{\xi}_t,$$

with

$$\boldsymbol{\eta}_t \sim N_K(\mathbf{0}, \mathbf{M}), \quad \boldsymbol{\xi}_t \sim N_n(\mathbf{0}, \sigma_\xi^2 \mathbf{I}_n), \quad \boldsymbol{\epsilon}_t \sim N_n(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I}_n).$$

In the present proposal, there is no separate measurement-error term in the standardized residual update, so the natural choice is

$$\sigma_\epsilon^2 = 0.$$

If a known measurement-error variance is desired in a specific application, it can be inserted without changing the formulas below.

4.2 Theorem 2 Plug-In Estimator

Define

$$\mathbf{H} = (\mathbf{F}'\mathbf{F})^{-1/2} \mathbf{F}' \mathbf{S} \mathbf{F} (\mathbf{F}'\mathbf{F})^{-1/2}.$$

Let the eigendecomposition of $\mathbf{H}$ be

$$\mathbf{H} = \mathbf{P} \text{diag}(d_1, \dots, d_K) \mathbf{P}',$$

where $d_1 \geq \dots \geq d_K \geq 0$ are the eigenvalues. Let $d_0 = 0$. Define L^* as the largest $L \in \{1, \dots, K\}$ such that

$$d_L > \max \left\{ \frac{1}{n-L} \left(\text{tr}(\mathbf{S}) - \sum_{k=1}^L d_k \right), \sigma_\epsilon^2 \right\},$$

if such an L exists, and let $L^* = 0$ otherwise.

Then the Tzeng–Huang estimators are

$$\hat{\sigma}_\xi^2 = \max \left\{ \frac{1}{n-L^*} \left(\text{tr}(\mathbf{S}) - \sum_{k=1}^{L^*} d_k \right) - \sigma_\epsilon^2, 0 \right\}, \quad (4)$$

and

$$\widehat{\mathbf{M}} = (\mathbf{F}'\mathbf{F})^{-1/2} \mathbf{P} \text{diag}(\hat{d}_1, \dots, \hat{d}_K) \mathbf{P}' (\mathbf{F}'\mathbf{F})^{-1/2}, \quad (5)$$

where

$$\hat{d}_k = \max(d_k - \hat{\sigma}_\xi^2 - \sigma_\epsilon^2, 0), \quad k = 1, \dots, K.$$

The updated low-rank covariance is then

$$\widehat{\mathbf{G}} = \mathbf{F} \widehat{\mathbf{M}} \mathbf{F}' + \hat{\sigma}_\xi^2 \mathbf{I}_n. \quad (6)$$

To preserve the Morris architecture, define

$$\widehat{\mathbf{A}} = \text{diag}(\widehat{\mathbf{G}}), \quad \widehat{\mathbf{R}} = \widehat{\mathbf{A}}^{-1/2} \widehat{\mathbf{G}} \widehat{\mathbf{A}}^{-1/2}. \quad (7)$$

4.3 Interpretation

The update in (4)–(7) does *not* treat the $K(K+1)/2$ free elements of $\mathbf{M}$ as a Bayesian parameter block requiring its own MCMC updates. Instead, it profiles the covariance block out by replacing $(\mathbf{M}, \sigma_\xi^2)$ with closed-form estimators computed from the current residual replicates. Thus the computational role of the MRTS layer is to reduce the dimension of the matrix decomposition from $n \times n$ to $K \times K$, rather than to introduce a large new stochastic parameter block.

5 Likelihood and Computation

5.1 Conditional Gaussian Likelihood

Given $(\beta, \lambda, \mathbf{z}_t, \mathbf{D}_t, \mathbf{R})$, the conditional distribution of $\mathbf{Y}_t$ is

$$\mathbf{Y}_t \mid \beta, \lambda, \mathbf{z}_t, \mathbf{D}_t, \mathbf{R} \sim N_n(\mathbf{X}_t\beta + \lambda\mathbf{D}_t|\mathbf{z}_t|, \mathbf{D}_t\mathbf{R}\mathbf{D}_t).$$

Equivalently, the standardized residuals satisfy

$$\mathbf{r}_t \mid \mathbf{R} \sim N_n(\mathbf{0}, \mathbf{R}), \quad t = 1, \dots, T.$$

Hence, up to an additive constant, the conditional log-likelihood of the residual block is

$$\ell(\mathbf{R} \mid \mathbf{r}_1, \dots, \mathbf{r}_T) = -\frac{T}{2} \log |\mathbf{R}| - \frac{1}{2} \sum_{t=1}^T \mathbf{r}_t' \mathbf{R}^{-1} \mathbf{r}_t. \quad (8)$$

5.2 Woodbury Identity for Fast Matrix Operations

Let

$$\mathbf{G} = \mathbf{F}\mathbf{M}\mathbf{F}' + \sigma_\xi^2 \mathbf{I}_n, \quad \sigma_\xi^2 > 0.$$

Then the Woodbury identity gives

$$\mathbf{G}^{-1} = \sigma_\xi^{-2} \mathbf{I}_n - \sigma_\xi^{-2} \mathbf{F} \left(\mathbf{M}^{-1} + \sigma_\xi^{-2} \mathbf{F}' \mathbf{F} \right)^{-1} \mathbf{F}' \sigma_\xi^{-2}. \quad (9)$$

Also, by the matrix determinant lemma,

$$\log |\mathbf{G}| = n \log(\sigma_\xi^2) + \log |\mathbf{M}| + \log \left| \mathbf{M}^{-1} + \sigma_\xi^{-2} \mathbf{F}' \mathbf{F} \right|. \quad (10)$$

Because

$$\mathbf{R} = \mathbf{A}^{-1/2} \mathbf{G} \mathbf{A}^{-1/2},$$

we obtain

$$\mathbf{R}^{-1} = \mathbf{A}^{1/2} \mathbf{G}^{-1} \mathbf{A}^{1/2}, \quad (11)$$

and

$$\log |\mathbf{R}| = \log |\mathbf{G}| - \sum_{i=1}^n \log(A_{ii}). \quad (12)$$

Thus all large-matrix computations needed in (8) can be reduced to operations involving $\mathbf{F}'\mathbf{F}$ and other $K \times K$ matrices, followed by matrix-vector products with $\mathbf{F}$.

5.3 Computational Complexity

For a single covariance update, the main costs are:

- forming $\mathbf{F}'\mathbf{r}_t$ for $t = 1, \dots, T$, which costs $O(nKT)$;
- forming the $K \times K$ matrix $\mathbf{F}'\mathbf{S}\mathbf{F} = T^{-1} \sum_{t=1}^T (\mathbf{F}'\mathbf{r}_t)(\mathbf{F}'\mathbf{r}_t)'$, which costs $O(K^2T)$ after the projected residuals are available;
- eigendecomposition or inversion of $K \times K$ matrices, which costs $O(K^3)$.

This replaces repeated direct inversions of dense $n \times n$ Matérn covariance matrices, whose dominant cost is typically $O(n^3)$ per update. When $K \ll n$, this is the main source of computational acceleration.

5.4 Identifiability Constraint Imposed by T

The computational gain from choosing $K \ll n$ does not remove the statistical constraint imposed by the number of replicates T . To see this, define the projected residual vectors

$$\tilde{\mathbf{r}}_t = (\mathbf{F}'\mathbf{F})^{-1/2}\mathbf{F}'\mathbf{r}_t \in \mathbb{R}^K, \quad t = 1, \dots, T.$$

Then the matrix whose eigenvalues drive the Tzeng–Huang update is

$$\mathbf{H} = (\mathbf{F}'\mathbf{F})^{-1/2}\mathbf{F}'\mathbf{S}\mathbf{F}(\mathbf{F}'\mathbf{F})^{-1/2} = \frac{1}{T} \sum_{t=1}^T \tilde{\mathbf{r}}_t \tilde{\mathbf{r}}_t'. \quad (13)$$

Therefore,

$$\text{rank}(\mathbf{H}) \leq \min(K, T). \quad (14)$$

If centered residuals are used in place of raw residual replicates, the bound is typically reduced to $\text{rank}(\mathbf{H}) \leq \min(K, T - 1)$.

Equation (14) implies that even when the replicates are conditionally independent and identically distributed under a common covariance matrix, the effective rank that can be learned for $\mathbf{M}$ is still limited by the sample size in time. In particular:

- if $K > T$, then at most T directions in the K -dimensional basis space can be identified from the replicate covariance;
- if K is close to T , then the leading eigendirections may still be estimable, but the smaller eigenvalues and associated eigenspaces are expected to be unstable;
- the theorem still provides a valid closed-form update, but the resulting $\widehat{\mathbf{M}}$ will be effectively low-rank and may be noisy in finite samples.

Consequently, basis selection should be guided not only by the spatial sample size n , but also by the number of temporal replicates T . A practical recommendation for the first implementation is to choose K so that $K \ll T$ in addition to $K \ll n$, and to regard the Tzeng–Huang truncation step as an adaptive effective-rank selector rather than as a license to let K grow without regard to T .

6 Hybrid Bayesian–Profile Inference Scheme

The proposed method is not a fully Bayesian model for $\mathbf{M}$. Instead, it is a Bayesian hierarchical model for the skew, scale, regression, and partition blocks, combined with a profiled low-rank covariance update. A practical algorithm is:

1. Update latent censored or missing values in $\mathbf{Y}_t$ as in the original Morris computation.
2. Update $(\boldsymbol{\beta}, \lambda)$ using their conditional Gaussian blocks.
3. Update the partition-specific latent variables $\mathbf{z}_t$, the scale terms $\mathbf{D}_t$, and any partition-location parameters exactly as in the original Morris sampler.
4. Compute residuals $\mathbf{r}_t$ using (3).
5. Form $\mathbf{S} = T^{-1} \sum_{t=1}^T \mathbf{r}_t \mathbf{r}_t'$.
6. Update $(\mathbf{M}, \sigma_\xi^2)$ by the closed-form formulas (4)–(5).
7. Construct $\mathbf{G}$, $\mathbf{A}$, and $\mathbf{R}$ via (6)–(7).
8. Evaluate quadratic forms and log-determinants using (9)–(12).

This scheme is best viewed as a hybrid MCMC/ECM algorithm: the Morris blocks are updated by Bayesian conditional simulation, whereas the low-rank covariance block is updated by a closed-form optimization step conditional on the current residuals.

6.1 Implementation Checklist

Before coding the first working version, the following implementation choices should be fixed:

1. **Fixed basis design.** The MRTS basis matrix $\mathbf{F}$ is treated as fixed throughout the analysis and is constructed outside the MCMC.
2. **Time-invariant covariance.** The residual replicates $\mathbf{r}_1, \dots, \mathbf{r}_T$ are assumed conditionally independent and identically distributed under a common spatial dependence matrix.
3. **Profiled covariance block.** The pair $(\mathbf{M}, \sigma_\xi^2)$ is updated by the closed-form Tzeng–Huang step rather than by a high-dimensional Bayesian parameter update.
4. **Correlation normalization.** The covariance $\mathbf{G}$ is converted to a correlation matrix $\mathbf{R}$ before entering the Morris likelihood, so that $\mathbf{D}_t$ remains responsible for marginal scale.
5. **Numerical stabilization.** The implementation should include a small ridge or floor when computing $(\mathbf{F}'\mathbf{F})^{-1/2}$ and when enforcing $\hat{\sigma}_\xi^2 \geq 0$.
6. **Basis size relative to T .** The chosen basis dimension K should respect the identifiability constraint described in (13)–(14).
7. **Validation targets.** The first implementation should be evaluated against the original Morris model using both runtime and predictive criteria, including uncertainty calibration and tail behavior.

6.2 Suggested Module Decomposition

A clean first implementation can be organized around the following modules:

1. `build_basis_matrix(...)`: construct the fixed MRTS basis matrix $\mathbf{F}$ at the observed locations, along with any stored quantities such as $\mathbf{F}'\mathbf{F}$.
2. `update_residuals(...)`: compute the conditional standardized residuals $\mathbf{r}_t = \mathbf{D}_t^{-1}(\mathbf{Y}_t - \mathbf{X}_t\boldsymbol{\beta} - \lambda\mathbf{D}_t|\mathbf{z}_t|)$.
3. `update_M_closed_form(...)`: compute $\mathbf{S}$, form $\mathbf{H}$, carry out the eigendecomposition, determine L^* , and return $(\widehat{\mathbf{M}}, \widehat{\sigma}_\xi^2)$.
4. `build_R_from_G(...)`: form $\widehat{\mathbf{G}} = \mathbf{F}\widehat{\mathbf{M}}\mathbf{F}' + \widehat{\sigma}_\xi^2\mathbf{I}_n$, compute $\widehat{\mathbf{A}} = \text{diag}(\widehat{\mathbf{G}})$, and return $\widehat{\mathbf{R}} = \widehat{\mathbf{A}}^{-1/2}\widehat{\mathbf{G}}\widehat{\mathbf{A}}^{-1/2}$.
5. `quadform_logdet(...)`: evaluate the quadratic forms and log-determinants needed in the residual likelihood using the Woodbury identity and matrix determinant lemma.
6. `update_morris_blocks(...)`: retain the existing Morris updates for censored data imputation, regression, skewness, partition-specific scale terms, and any knot-location parameters.
7. `predict_at_new_sites(...)`: construct basis values at prediction sites and produce posterior predictive draws using the fitted low-rank spatial layer and the current Morris mean/scale components.

This modular decomposition is intended to make the first implementation easy to debug: the low-rank covariance block can be tested first in a simplified Gaussian setting before it is reattached to the full censored skew- t sampler.

7 Optional One-Parameter Bayesian Extension

If some uncertainty in the covariance block is desired without returning to a full $K(K+1)/2$ -dimensional MCMC update, one simple extension is to introduce a scalar expansion parameter $c > 0$ and set

$$\mathbf{M} = c\widehat{\mathbf{M}},$$

where $\widehat{\mathbf{M}}$ is the Tzeng–Huang plug-in estimator. One may then place a prior on c , for example

$$c \sim \text{Gamma}(a_c, b_c),$$

and update only c within the MCMC. This preserves the dominant eigenstructure estimated from the closed-form step while allowing limited posterior uncertainty in the overall strength of the low-rank dependence.

8 Modeling Assumptions and Practical Remarks

1. The MRTS basis matrix $\mathbf{F}$ is treated as fixed and prespecified.
2. The Tzeng–Huang update relies on replicate residual vectors $\mathbf{r}_1, \dots, \mathbf{r}_T$ that are conditionally independent with a common spatial dependence structure.

3. In the present proposal, σ_ϵ^2 is set to zero because the standardized residual vectors are treated as directly observed for the purpose of the covariance update. If an external measurement-error variance is known, it can be inserted into the update formulas.
4. Standardizing $\mathbf{G}$ to $\mathbf{R}$ is a deliberate modeling choice made to preserve the original Morris interpretation that $\mathbf{D}_t$ controls marginal scale while the spatial layer controls dependence.
5. The basis dimension K should remain well below n , but it must also be chosen with care relative to T because the effective rank learnable from the replicate covariance is constrained by (14).

9 Summary

The proposed model keeps the Morris skew- t hierarchy intact at the observation level,

$$\mathbf{Y}_t = \mathbf{X}_t \boldsymbol{\beta} + \lambda \mathbf{D}_t |\mathbf{z}_t| + \mathbf{D}_t \mathbf{v}_t,$$

but replaces the site-level Matérn dependence with an MRTS low-rank spatial layer. The unknown covariance matrix $\mathbf{M}$ is not sampled as a large Bayesian parameter block. Instead, it is updated in closed form from current residuals using the Tzeng–Huang estimator, after which the implied covariance matrix is standardized to a correlation matrix and inserted back into the Morris hierarchy. This retains computational scalability while preserving the original interpretation of the skew and scale blocks.